

# Permutations, Inversions, and Determinants

---

N-20220725

## §1.1 Inversions and signs

The note begins with inversions of a permutation. For a permutation

$$\sigma = (\sigma_1, \dots, \sigma_n),$$

an inversion is a pair  $i < j$  such that

$$\sigma_i > \sigma_j.$$

The number of inversions is denoted  $\tau(\sigma)$ . The sign of the permutation is

$$\text{sgn}(\sigma) = (-1)^{\tau(\sigma)}.$$

For example, for 426315, the inversions are counted by checking how many later entries are smaller than each entry. The note obtains

$$\tau(426315) = 8.$$

In determinant expansions, the sign of each term is exactly the sign of a permutation. Thus a term like

$$a_{21}a_{53}a_{16}a_{42}a_{65}a_{34}$$

corresponds to a row/column permutation, and its sign is determined by its inversion number.

## §1.2 Definition of determinant

**Definition 1.2.1** (Determinant). The determinant is the alternating multilinear function of the columns of a square matrix normalized by

$$\det I = 1.$$

It measures oriented volume scaling.

The determinant of an  $n \times n$  matrix  $A = (a_{ij})$  is

$$\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}.$$

This formula is not meant for computation in large dimensions. Its value is conceptual: it explains why row swaps change sign, why repeated rows give zero, and why the determinant is multilinear in rows and columns.

### §1.3 Basic determinant properties

The note records the following standard properties:

- (i) Transposition does not change the determinant:

$$\det A^T = \det A.$$

- (ii) Swapping two rows changes the sign.
- (iii) Pulling a scalar out of one row pulls it out of the determinant.
- (iv) The determinant is additive in each row separately.
- (v) Adding a multiple of one row to another row does not change the determinant.

From these one immediately gets the zero properties:

- (i) if a row is zero, then the determinant is zero;
- (ii) if two rows are proportional, then the determinant is zero.

### §1.4 Sarrus and triangular matrices

For a  $3 \times 3$  matrix, Sarrus' rule gives a convenient visual computation. But the note also emphasizes that triangular matrices are easier: if  $A$  is upper or lower triangular, then

$$\det A = a_{11}a_{22} \cdots a_{nn}.$$

This is why row reduction is so powerful. We use row operations to simplify a determinant into triangular form while tracking sign changes and scalar factors.

### §1.5 Cofactor expansion

The cofactor of  $a_{ij}$  is

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where  $M_{ij}$  is the determinant obtained by deleting row  $i$  and column  $j$ . Expanding along row  $i$ ,

$$\det A = \sum_{j=1}^n a_{ij} C_{ij}.$$

A page in the note works through determinant computations by choosing rows or columns with many zeros. This is the main practical advice: do not expand randomly; expand where the determinant is sparse.

### §1.6 A structured determinant

A common determinant in the note has the form

$$D_n = \begin{vmatrix} x & a & a & \cdots & a \\ a & x & a & \cdots & a \\ \vdots & \vdots & \vdots & & \vdots \\ a & a & a & \cdots & x \end{vmatrix}.$$

This matrix can be written

$$D_n = (x - a)I + aJ,$$

where  $J$  is the all-one matrix. The vector  $(1, \dots, 1)$  is an eigenvector with eigenvalue

$$x + (n - 1)a,$$

and any vector whose coordinates sum to 0 has eigenvalue

$$x - a.$$

Therefore

$$D_n = (x + (n - 1)a)(x - a)^{n-1}.$$

The note obtains this by row and column operations; the eigenvalue argument is the higher-level explanation.

## §1.7 The determinant line and exterior powers

The previous computational rules all point to one structure. Let  $V$  be an  $n$ -dimensional vector space. The top exterior power

$$\Lambda^n V$$

is one-dimensional and is called the determinant line. A nonzero element of this line is a volume form; a choice up to positive scalar is an orientation.

**Definition 1.7.1** (Determinant via the top exterior power). For a linear map  $A : V \rightarrow V$ , the induced map

$$\Lambda^n A : \Lambda^n V \rightarrow \Lambda^n V$$

acts on a one-dimensional space, hence is multiplication by a scalar. That scalar is  $\det A$ .

In a basis  $e_1, \dots, e_n$ , if the columns of  $A$  are  $v_1, \dots, v_n$ , then

$$v_1 \wedge \dots \wedge v_n = (\det A)e_1 \wedge \dots \wedge e_n.$$

This single identity explains the main determinant rules:

- (i) multilinearity comes from multilinearity of the wedge product;
- (ii) repeated or dependent columns wedge to zero;
- (iii) swapping two columns changes the sign;
- (iv) multiplicativity follows from  $\Lambda^n(AB) = (\Lambda^n A)(\Lambda^n B)$ .

## §1.8 Why permutation signs are forced

The sign of a permutation is not an arbitrary convention. It is the unique homomorphism

$$S_n \longrightarrow \{\pm 1\}$$

that sends every transposition to  $-1$ . The inversion number gives a concrete formula:

$$\text{sgn}(\sigma) = (-1)^{\# \text{ inversions of } \sigma}.$$

Because the determinant is alternating, permuting the columns by  $\sigma$  must multiply the volume form by this sign. This is why the Leibniz formula has exactly the sign pattern it has.

## §1.9 Minors as exterior coordinates

The same exterior-power viewpoint also explains minors. If  $A : V \rightarrow W$ , then

$$\Lambda^k A : \Lambda^k V \rightarrow \Lambda^k W$$

is represented, in the induced exterior bases, by the  $k \times k$  minors of  $A$ . Therefore

$$\text{rank } A < k \iff \Lambda^k A = 0 \iff \text{all } k \times k \text{ minors vanish.}$$

Thus minors are not arbitrary subdeterminants; they are coordinates of an exterior-power map.

## §1.10 Cramer's rule from volume ratios

Cramer's rule replaces one column of  $A$  by  $b$ . Geometrically, solving  $Ax = b$  asks for the coordinates of  $b$  in the parallelepiped basis given by the columns of  $A$ . The coordinate  $x_i$  is the ratio of two oriented volumes:

$$x_i = \frac{\det A_i(b)}{\det A}.$$

This interpretation makes the formula less mysterious: a coordinate is measured by replacing the corresponding basis vector and comparing volumes.